

Клонирую исходники ядра - ванильную ветку Торвальдса (git), сразу на стабильный тег v6.17.

```
git clone --depth 1 --branch v6.17 https://git.kernel.org/pub/scm/linux/kernel/git/torvalds/linux.git linux
```

```
nm@home:~$ git clone --depth 1 --branch v6.17 https://git.kernel.org/pub/scm/linux/kernel/git/torvalds/linux.git linux
Updating files: 100% (90506/90506)
Updating files: 100% (90506/90506), done.
```

Создаю конфигурацию под свою архитектуру - make defconfig (за основу берётся x86_64_defconfig).

```
cd linux && make defconfig
```

```
nm@home:~$ make defconfig
*** Default configuration is based on 'x86_64_defconfig'
#
# configuration written to .config
#
```

Чтобы ядро точно поднялось на моём железе, беру за основу конфиг рабочего ядра и отключаю подписи Canonical (иначе сборка падает на сертификатах).

```
cp /boot/config-$(uname -r) .config && make olddefconfig
scripts/config --disable SYSTEM_TRUSTED_KEYS
scripts/config --disable SYSTEM_REVOCATION_KEYS
scripts/config --disable GENDWARFSYMS
```

```
nm@home:~$ cp /boot/config-$(uname -r) .config && make olddefconfig
(.config <- конфиг рабочего ядра 6.17.0)
#
# No change to .config
#
nm@home:~$ scripts/config --disable SYSTEM_TRUSTED_KEYS
ok
nm@home:~$ scripts/config --disable SYSTEM_REVOCATION_KEYS
ok
nm@home:~$ scripts/config --disable GENDWARFSYMS
ok
```

Собираю ядро и модули параллельно по числу ядер. На выходе - образ bzImage.

```
make -j$(nproc) && make -j$(nproc) modules
```

```
CC      drivers/iommu/iommu-traces.o
CC      drivers/acpi/acpica/nsutils.o
CC      arch/x86/kernel/kvmclock.o
AR      fs/proc/built-in.a
CC      fs/sysfs/file.o
CC      net/xfrm/xfrm_hash.o
CC      lib/fw_table.o
CC      kernel/cgroup/namespace.o
CC      kernel/utsname.o
CC      drivers/acpi/acpica/nswalk.o
CC      net/core/hotdata.o
CC      net/ipv4/tcp.o
CC      arch/x86/kernel/paravirt.o
CC      net/ethtool/fec.o
CC      net/xfrm/xfrm_input.o
CC      drivers/acpi/acpica/nsxfeval.o
CC      fs/sysfs/dir.o
CC      net/core/netdev_rx_queue.o
CC      drivers/iommu/iommu-sysfs.o
AR      lib/lib.a
AR      lib/built-in.a
CC      drivers/acpi/fan_hwmon.o
CC      kernel/cgroup/cgroup-v1.o
CC      net/netfilter/nf_conntrack_proto_icmp.o
CC      drivers/acpi/acpica/nsxfname.o
CC      fs/sysfs/symlink.o
CC      fs/sysfs/mount.o
CC      arch/x86/kernel/pvclock.o
CC      drivers/iommu/dma-iommu.o
CC      drivers/acpi/acpica/nsxfobj.o
CC      net/core/net-procfs.o
CC      net/ethtool/eeprom.o
CC      net/unix/af_unix.o
CC      drivers/acpi/acpica/psargs.o
CC      kernel/cgroup/freezer.o
CC      arch/x86/kernel/pcspeaker.o
CC      net/netfilter/nf_conntrack_extend.o
CC      fs/sysfs/group.o
CC      net/xfrm/xfrm_output.o
CC      drivers/acpi/acpica/psloop.o
CC      arch/x86/kernel/check.o
CC      net/ethtool/stats.o
```

Устанавливаю собранное ядро в систему: модули в /lib/modules, ядро в /boot, GRUB добавляет новый пункт.

```
sudo make modules_install && sudo make install
```

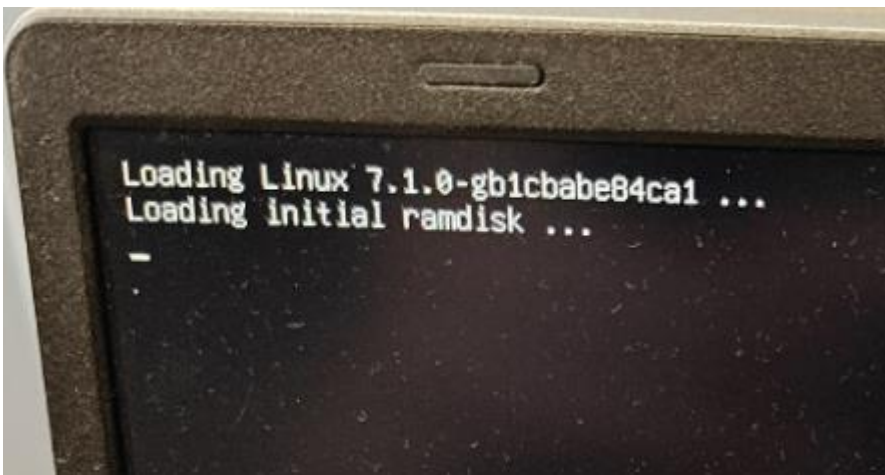
```

INSTALL /lib/modules/6.17.0/kernel/net/openvswitch/vport-gre.ko
INSTALL /lib/modules/6.17.0/kernel/net/vmw_vsock/vsock.ko
INSTALL /lib/modules/6.17.0/kernel/net/vmw_vsock/vsock_diag.ko
INSTALL /lib/modules/6.17.0/kernel/net/vmw_vsock/vmw_vsock_vmci_transport.ko
INSTALL /lib/modules/6.17.0/kernel/net/vmw_vsock/vmw_vsock_virtio_transport.ko
INSTALL /lib/modules/6.17.0/kernel/net/vmw_vsock/vmw_vsock_virtio_transport_common.ko
INSTALL /lib/modules/6.17.0/kernel/net/vmw_vsock/hv_sock.ko
INSTALL /lib/modules/6.17.0/kernel/net/vmw_vsock/vsock_loopback.ko
INSTALL /lib/modules/6.17.0/kernel/net/nsh/nsh.ko
INSTALL /lib/modules/6.17.0/kernel/net/hsr/hsr.ko
INSTALL /lib/modules/6.17.0/kernel/net/qrtr/qrtr.ko
INSTALL /lib/modules/6.17.0/kernel/net/qrtr/qrtr-smd.ko
INSTALL /lib/modules/6.17.0/kernel/net/qrtr/qrtr-tun.ko
INSTALL /lib/modules/6.17.0/kernel/net/qrtr/qrtr-mhi.ko
INSTALL /lib/modules/6.17.0/kernel/virt/lib/irqbypass.ko
DEPMOD /lib/modules/6.17.0
INSTALL /boot
run-parts: executing /etc/kernel/postinst.d/initramfs-tools 6.17.0 /boot/vmlinuz-6.17.0
update-initramfs: Generating /boot/initrd.img-6.17.0
run-parts: executing /etc/kernel/postinst.d/unattended-upgrades 6.17.0 /boot/vmlinuz-6.17.0
run-parts: executing /etc/kernel/postinst.d/update-notifier 6.17.0 /boot/vmlinuz-6.17.0
run-parts: executing /etc/kernel/postinst.d/xx-update-initrd-links 6.17.0 /boot/vmlinuz-6.17.0
run-parts: executing /etc/kernel/postinst.d/zz-shim 6.17.0 /boot/vmlinuz-6.17.0
run-parts: executing /etc/kernel/postinst.d/zz-update-grub 6.17.0 /boot/vmlinuz-6.17.0
Sourcing file `/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-7.1.0-gb1cbabe84ca1
Found initrd image: /boot/initrd.img-7.1.0-gb1cbabe84ca1
Found linux image: /boot/vmlinuz-6.17.0-35-generic
Found initrd image: /boot/initrd.img-6.17.0-35-generic
Found linux image: /boot/vmlinuz-6.17.0-29-generic
Found initrd image: /boot/initrd.img-6.17.0-29-generic
Found linux image: /boot/vmlinuz-6.17.0
Found initrd image: /boot/initrd.img-6.17.0
Found linux image: /boot/vmlinuz-6.17.0.old
Found initrd image: /boot/initrd.img-6.17.0
Found memtest86+ 64bit EFI image: /boot/memtest86+x64.efi
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done

```

Перезагрузился, выбрал своё ядро в GRUB и загрузился на реальном железе. Проверка: `uname 6.17.0`, `systemd-detect-virt = none` (железо), `/proc/version` - моя сборка.

(проверял еще 7.1.0 – моя самая первая сборка, она не запустилась)



```
uname -a; systemd-detect-virt; cat /proc/version
```

```
nm@home:~$ uname -a; systemd-detect-virt; cat /proc/version
Linux nm-Aspire-AL15-42P 6.17.0 #2 SMP PREEMPT_DYNAMIC Tue Jun 16 20:30:47 +07 2026 x86_64 x86_64 x86_64 GNU/Linux
none
Linux version 6.17.0 (nm@nm-Aspire-AL15-42P) (gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #2 SMP PREEMPT_DYNAMIC Tue Jun 16 20:30:47 +07 2026
```

То же ядро собрал и под ARM кросс-компилятором - образ zImage и device tree.

```
make ARCH=arm multi_v7_defconfig && make ARCH=arm CROSS_COMPILE=arm-linux-gnueabihf- -j$(nproc) zImage dtbs
```

```

#
# configuration written to .config
#
SYNC    include/config/auto.conf.cmd
*
* Restart config...
*
* Kernel Features
*
Symmetric Multi-Processing (SMP) [Y/n/?] y
  Allow booting SMP kernel on uniprocessor systems (SMP_ON_UP) [Y/n/?] y
Support cpu topology definition (ARM_CPU_TOPOLOGY) [Y/n/?] y
  Multi-core scheduler support (SCHED_MC) [N/y/?] n
  SMT scheduler support (SCHED_SMT) [N/y/?] n
Architected timer support (HAVE_ARM_ARCH_TIMER) [Y/?] y
Multi-Cluster Power Management (MCPM) [Y/?] y
big.LITTLE support (Experimental) (BIG_LITTLE) [N/y/?] n
Memory split
> 1. 3G/1G user/kernel split (VMSPLIT_3G)
  2. 3G/1G user/kernel split (for full 1G low memory) (VMSPLIT_3G_OPT)
  3. 2G/2G user/kernel split (VMSPLIT_2G)
  4. 1G/3G user/kernel split (VMSPLIT_1G)
choice[1-4?]: 1
Maximum number of CPUs (2-32) (NR_CPUS) [16] 16
Support for hot-pluggable CPUs (HOTPLUG_CPU) [Y/?] y
Support for the ARM Power State Coordination Interface (PSCI) (ARM_PSCI) [Y/?] y
Timer frequency
> 1. 100 Hz (HZ_100)
  2. 200 Hz (HZ_200)
  3. 250 Hz (HZ_250)
  4. 300 Hz (HZ_300)
  5. 500 Hz (HZ_500)
  6. 1000 Hz (HZ_1000)
choice[1-6?]: 1
Compile the kernel in Thumb-2 mode (THUMB2_KERNEL) [N/y/?] n
Runtime patch udiv/sdiv instructions into __aeabi_{u}div() (ARM_PATCH_IDIV) [Y/n/?] y
Use the ARM EABI to compile the kernel (AEABI) [Y/?] y
  Allow old ABI binaries to run with this kernel (EXPERIMENTAL) (OABI_COMPAT) [N/y/?] n
High Memory Support (HIGHMEM) [Y/?] y
  Allocate 2nd-level pagetables from highmem (HIGHPT64) [Y/n/?] y
Enable privileged no-access (ARM_PAN) [Y/n/?] y
Use PLTs to allow module memory to spill over into vmalloc area (ARM_MODULE_PLTS) [Y/n/?] y
Order of maximal physically contiguous allocations (ARCH_FORCE_MAX_ORDER) [11] 11
Use kernel mem{cpy,set}() for {copy_to,clear}_user() (UACCESS_WITH_MEMCPY) [N/y/?] n
Enable paravirtualization code (PARAVIRT) [N/y/?] n
Paravirtual steal time accounting (PARAVIRT_TIME_ACCOUNTING) [N/y/?] n
Xen guest support on ARM (XEN) [N/y/?] n
Use a unique stack canary value for each task (STACKPROTECTOR_PER_TASK) [Y/n/?] (NEW) y
Use a unique stack canary value for each task (STACKPROTECTOR_PER_TASK) [Y/n/?] (NEW) y
  CALL    scripts/checksyscalls.sh
  Kernel: arch/arm/boot/Image is ready
  Kernel: arch/arm/boot/zImage is ready

```

Запускаю ARM-ядро в QEMU без корневой ФС - kernel panic (init не найден), как и должно быть.

```
qemu-system-arm -M vexpress-a9 -nographic -kernel arch/arm/boot/zImage -dtb  
arch/arm/boot/dts/arm/vexpress-v2p-ca9.dtb -append "console=ttyAMA0"
```

```
[ 2.545261] (driver?)  
[ 2.545268] 0106          65536 ram6  
[ 2.545274] (driver?)  
[ 2.545280] 0107          65536 ram7  
[ 2.545286] (driver?)  
[ 2.545292] 0108          65536 ram8  
[ 2.545298] (driver?)  
[ 2.545305] 0109          65536 ram9  
[ 2.545310] (driver?)  
[ 2.545317] 010a          65536 ram10  
[ 2.545323] (driver?)  
[ 2.545329] 010b          65536 ram11  
[ 2.545335] (driver?)  
[ 2.545341] 010c          65536 ram12  
[ 2.545347] (driver?)  
[ 2.545353] 010d          65536 ram13  
[ 2.545359] (driver?)  
[ 2.545366] 010e          65536 ram14  
[ 2.545371] (driver?)  
[ 2.545378] 010f          65536 ram15  
[ 2.545384] (driver?)  
[ 2.545400] 1f00          131072 mtdblock0  
[ 2.545411] (driver?)  
[ 2.545462] List of all bdev filesystems:  
[ 2.545474] ext3  
[ 2.545485] ext2  
[ 2.545489] ext4  
[ 2.545493] squashfs  
[ 2.545497] vfat  
[ 2.545500] msdos  
[ 2.545505] ntfs  
[ 2.545508] ntfs3  
[ 2.545515]  
[ 2.545646] Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,0)  
[ 2.551025] CPU: 0 UID: 0 PID: 1 Comm: swapper/0 Not tainted 6.17.0 #1 NONE  
[ 2.551299] Hardware name: ARM-Versatile Express  
[ 2.551529] Call trace:  
[ 2.551902] unwind_backtrace from show_stack+0x10/0x14  
[ 2.552592] show_stack from dump_stack_lvl+0x54/0x68  
[ 2.552756] dump_stack_lvl from vpanic+0xf4/0x304  
[ 2.552913] vpanic from __do_trace_suspend_resume+0x0/0x4c  
[ 2.553576] ---[ end Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,0) ]---
```